Parametric and Semiparametric Copula Model Selection

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Abstract

Selecting the most appropriate copula model is a fundamental challenge in multivariate dependence modeling. This project investigates both parametric and semiparametric approaches to copula model selection by replicating the simulation study from Ko and Hjort (2019) and a little extending the simulations from Jordanger and Tjøstheim (2014). First, in the fully parametric setting under two-stage maximum likelihood estimation, we replicate simulations comparing AIC based on two-stage MLE ($\mathrm{AIC}_{\mathrm{2MLE}}$) against the model-robust Copula Information Criterion derived for two-stage MLE ($\mathrm{CIC}_{\mathrm{2MLE}}$). Second, we examine the semiparametric setting, where marginals are estimated empirically. In this context, the naive adaptation of AIC to pseudo-likelihood (${}^{p}\mathrm{AIC}$) lacks theoretical justification, while valid alternatives like leave-one-out cross-validation (${}^{p}\mathrm{xv}_1$) and its approximation ($\mathrm{xv}_{\mathrm{CIC}}$) are computationally intensive and offer little practical advantage over ${}^{p}\mathrm{AIC}$. We also propose and evaluate a K_n-fold cross-validation criterion. By utilizing data-driven choices for the number of folds K_n, we aim to mimic the theoretical validity of ${}^{p}\mathrm{xv}_1$ while alleviating its severe computational burden. Through an extensive simulation study, we benchmark these criteria against the ${}^{p}\mathrm{AIC}$; however, none of the considered criteria outperform the ${}^{p}\mathrm{AIC}$.

1 Introduction

Modeling multivariate data requires an understanding of the underlying dependence structure between variables. Consequently, copulas are widely used in applications across finance, insurance, and hydrology because they allow for the separation of marginal behavior from the dependence structure. Formally, for a dimension $d \in \mathbb{N}$, a function $C : [0,1]^d \to [0,1]$ is called a copula if it is a joint cumulative distribution function (cdf) with uniform marginals on $[0,1]$. According to Sklar's Theorem, any multivariate cdf F can be decomposed into its marginal cdfs $F_1, \ldots, F_d$ and a copula C, such that:

$$F(x_1, \ldots, x_d) = C(F_1(x_1), \ldots, F_d(x_d)).$$

If we assume that F is the cdf corresponding to an absolutely continuous distribution, then the copula C is uniquely defined. In this case, Sklar's Theorem can be expressed in terms of probability density functions (pdfs) as:

$$f(x_1, \ldots, x_d) = c(F_1(x_1), \ldots, F_d(x_d)) \prod_{l=1}^{d} f_l(x_l),$$

where

- $f(x_1, \ldots, x_d) = \partial^d F(x_1, \ldots, x_d)/\partial x_1 \ldots \partial x_d$ is the joint pdf corresponding to F,
- $c(u_1, \ldots, u_d) = \partial^d C(u_1, \ldots, u_d)/\partial u_1 \ldots \partial u_d$ is the copula pdf corresponding to C,
- $f_l(x_l) = \partial F_l(x_l)/\partial x_l$ is the marginal pdf corresponding to F_l for $l = 1, \ldots, d$.

Throughout this project, we will focus exclusively on absolutely continuous distributions. Sklar's Theorem allows for the construction of multivariate models by combining marginal models with a copula model. A natural practical question that arises is the selection of the most appropriate copula. That is, assuming we have fitted a set of candidate copula families (e.g., Clayton, Gumbel, Frank, Gaussian), how do we select the model that best captures the underlying dependence in our data? This project explores the problem of copula model selection through two distinct methodologies: fully parametric and semiparametric approaches. By replicating and a little extending key findings from the literature, we aim to evaluate established information criteria.

1.1 The Fully Parametric Approach and Two-Stage MLE

In a fully parametric setting, both the marginal distributions and the copula are assumed to have a parametric form. Formally, this means that the copula C can be parameterized by a real vector $\boldsymbol{\theta}$, and each marginal cdf F_l can be parameterized by a real vector $\boldsymbol{\gamma}_l$ for $l = 1, \ldots, d$. By Sklar's theorem, the joint density f can then be expressed as

$$f(x_1, \ldots, x_d; \boldsymbol{\theta}, \boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_d) = c(F_1(x_1; \boldsymbol{\gamma}_1), \ldots, F_d(x_d; \boldsymbol{\gamma}_d); \boldsymbol{\theta}) \prod_{l=1}^{d} f_l(x_l; \boldsymbol{\gamma}_l). \tag{1}$$

Assuming we observe a multivariate data sample $\mathcal{X}_n$ from an unknown data-generating pdf f°, let us consider $B \in \mathbb{N}$ possible parametric families:

$$\mathcal{F}_b = \left\{ f(\cdot; \boldsymbol{\theta}_b, \boldsymbol{\gamma}_{b,1}, \ldots, \boldsymbol{\gamma}_{b,d}) : (\boldsymbol{\theta}_b, \boldsymbol{\gamma}_{b,1}, \ldots, \boldsymbol{\gamma}_{b,d}) \in \Theta_b \times \Gamma_{b,1} \ldots \times \Gamma_{b,d} \right\} \quad \text{for } b = 1, \ldots, B.$$

One can then simultaneously estimate all model parameters $(\boldsymbol{\theta}, \boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_d)$ using maximum likelihood estimation (MLE). Thus, by fitting each family $\mathcal{F}_b$ to the data $\mathcal{X}_n$, one obtains the maximum likelihood estimates $\left(\widehat{\boldsymbol{\theta}}_b, \widehat{\boldsymbol{\gamma}}_{b,1}, \ldots, \widehat{\boldsymbol{\gamma}}_{b,d}\right)$ for $b = 1, \ldots, B$. Our goal is to rank the fitted models to identify the most suitable one. To achieve this, classical criteria such as $\mathrm{AIC}_{\mathrm{MLE}}$ or $\mathrm{TIC}_{\mathrm{MLE}}$ can be applied, depending on whether the true density f° belongs to $\mathcal{F}_b$ or not. However, the joint estimation of all parameters $(\boldsymbol{\theta}, \boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_d)$ is often computationally expensive and numerically unstable in high dimensions. As an alternative, the two-stage maximum likelihood method, which is also called as Inference Functions for Margins, is widely used. In the first stage, the marginal parameters $\boldsymbol{\gamma}_l$ for $l = 1, \ldots, d$ are estimated by MLE. In the second stage, these estimates are plugged into the joint likelihood to estimate the copula parameter $\boldsymbol{\theta}$, again by MLE. However, it is important to note that traditional model selection criteria, such as $\mathrm{AIC}_{\mathrm{MLE}}$ and $\mathrm{TIC}_{\mathrm{MLE}}$, are theoretically valid only under full MLE, and not under two-stage MLE. To address this, Ko and Hjort (2019) developed the Copula Information Criterion ($\mathrm{CIC}_{\mathrm{2MLE}}$) specifically for two-stage MLE. The $\mathrm{CIC}_{\mathrm{2MLE}}$ is a model-robust criterion analogous to the $\mathrm{TIC}_{\mathrm{MLE}}$; it aims to select the model that minimizes the Kullback-Leibler (KL) divergence from the true data-generating mechanism f° without assuming the true model is among the candidates $\mathcal{F}_b$. If the strong assumption is made that the candidate model is correctly specified, the $\mathrm{CIC}_{\mathrm{2MLE}}$ simplifies to $\mathrm{AIC}_{\mathrm{2MLE}}$. Note that we denote the AIC under two-stage MLE as $\mathrm{AIC}_{\mathrm{2MLE}}$ to distinguish it from MLE based $\mathrm{AIC}_{\mathrm{MLE}}$. While $\mathrm{CIC}_{\mathrm{2MLE}}$ conceptually improves upon $\mathrm{AIC}_{\mathrm{2MLE}}$ by penalizing misspecified models more heavily through its bias-correction term, simulations show that as sample sizes grow, the log-likelihood term dominates, resulting in minimal practical differences between the two criteria. Furthermore, the calculation of the bias-correction term for $\mathrm{CIC}_{\mathrm{2MLE}}$ is computationally expensive. To the best of my knowledge, the development of numerical techniques to mitigate this computational cost remains an open question. In this project, we independently replicate the simulation study from Ko and Hjort (2019) to confirm the behavior of $\mathrm{AIC}_{\mathrm{2MLE}}$ and $\mathrm{CIC}_{\mathrm{2MLE}}$.

1.2 The Semiparametric Approach and MPLE

One of the possible drawbacks of the fully parametric approach discussed above is that the estimates of the dependence parameters $\boldsymbol{\theta}$ are highly sensitive to the choice of the marginal models. In other words, the misspecification of the true marginal distributions can severely distort the estimates of the copula parameters $\boldsymbol{\theta}$. Therefore, in cases where the true nature of the marginals is unknown, it may be preferable to leave the margins unspecified. In the semiparametric approach, the marginal distributions are estimated empirically using scaled empirical cumulative distribution functions (ecdfs), transforming the data into so-called pseudo-observations. The copula parameters are then estimated via maximum pseudo-likelihood estimation (MPLE). In this setting, we want to fit several parametric copula models while keeping the marginals unspecified, and then select the best copula model. Many investigations, such as Chen and Fan (2005) and McNeil et al. (2005) (Chapter 5), used a naive adaptation of the AIC to the semiparametric case as a model selection criterion:

$$^{p}\mathrm{AIC} = 2 \cdot \left[{}^{p}\ell\left({}^{p}\widehat{\boldsymbol{\theta}}\right) - \dim(\boldsymbol{\theta}) \right],$$

where ${}^{p}\ell(\boldsymbol{\theta})$ is the pseudo-log-likelihood (which will be formally defined in Section 3), and ${}^{p}\widehat{\boldsymbol{\theta}} = \operatorname{argmax}\ {}^{p}\ell(\boldsymbol{\theta})$ is the MPLE. When using the ${}^{p}\mathrm{AIC}$, it was implicitly hoped that because the MPLE heuristically resembled the MLE, there would be a continuous relationship between the two techniques model selection behaviors, making the ${}^{p}\mathrm{AIC}$ formula approximately valid. However, the problem with the above formula is that it ignores the noise introduced in the transformation step, when the data are transformed into pseudo-observations.

In the paper Grønneberg and Hjort (2014), the authors adapted the arguments from the classical $\mathrm{AIC}_{\mathrm{MLE}}$ formula to the semiparametric case. First, they looked at copula model selection from a loss-function perspective, trying to minimize a certain Kullback-Leibler information distance to the true copula model. Under very restrictive conditions, they derived an information criterion which is very different from classical $\mathrm{AIC}_{\mathrm{MLE}}$, and for this criterion, the formula ${}^{p}\mathrm{AIC}$ is approximately valid. However, the problem is that the conditions that they impose are not satisfied for the most popular copula models, which means that ${}^{p}\mathrm{AIC}$ is not theoretically valid for the widely used copula families. In a nutshell, the key problem is that for the most popular copula models like Clayton, Gumbel, Gaussian, and others, the density of those copulas grows very fast to infinity near the edges of the unit hypercube $[0,1]^d$. As a possible alternative, the authors looked at copula model selection from a prediction perspective, and proposed the leave-one-out cross-validation information criterion ${}^{p}\mathrm{xv}_1$ and its computationally simpler approximation $\mathrm{xv}_{\mathrm{CIC}}$. These criteria, on the other hand, can be applied to all common copula models.

Paradoxically, despite the rigorous theoretical foundation of $\mathrm{xv}_{\mathrm{CIC}}$, a simulation study by Jordanger and Tjøstheim (2014) revealed only minor differences in practical performance between theoretically valid $\mathrm{xv}_{\mathrm{CIC}}$ and theoretically unjustified ${}^{p}\mathrm{AIC}$. From my personal (heuristic) understanding, I believe this might be because $\mathrm{xv}_{\mathrm{CIC}}$ was derived from a prediction perspective, but problems involving copula models are usually unsupervised. In other words, these problems often focus on fitting rather than predicting.

In this project, we independently replicate the simulation study from Jordanger and Tjøstheim (2014) to confirm the behavior of $\mathrm{xv}_{\mathrm{CIC}}$ and ${}^{p}\mathrm{AIC}$. We extend their work by incorporating the computationally intensive leave-one-out cross-validation criterion ${}^{p}\mathrm{xv}_1$ to compare its performance against the aforementioned criteria. Furthermore, as an exploratory extension, we propose a K_n-fold cross-validation criterion xv_{K_n} that utilizes a data-dependent choice for the number of folds, K_n. The motivation for this approach stems from the fact that ${}^{p}\mathrm{xv}_1$ is a special case of K_n-fold cross-validation (where $K_n = n$ and the validation set size is $n_v = 1$) that satisfies the asymptotic property $n_v/n = 1/n \underset{n\to\infty}{\longrightarrow} 0$. To explore xv_{K_n}, we first examine the behavior of K_n-fold cross-validation using common fixed choices for the number of folds, specifically $K_n \in \{2, 5, 10\}$. We then evaluate data-dependent values, setting

$$K_n \in \left\{ \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\}.$$

These specific choices are selected because they preserve the property $n_v/n = 1/K_n \underset{n\to\infty}{\longrightarrow} 0$, thereby mimicking the theoretical valid criterion ${}^{p}\mathrm{xv}_1$.

2 Evaluating Information Criteria under Two-Stage MLE

In the fully parametric framework, model selection aims to identify the specific combination of marginal distributions and a copula function that best approximates the true, unknown data-generating density f°. As highlighted in the introduction, estimating all parameters simultaneously using full MLE is often computationally intractable for high-dimensional data. Therefore, a widely used alternative is two-stage MLE.

2.1 Two-Stage Maximum Likelihood Estimation

The technical setting is as follows. Let $\boldsymbol{X} = (X_1, \ldots, X_d)^T$ be a d-variate continuous random vector with a true joint density $f^\circ(x_1, \ldots, x_d)$ and let $\boldsymbol{X}_i = (X_{i,1}, \ldots, X_{i,d})^T$ for $i = 1, \ldots, n$ be independent realizations of this vector. The true joint pdf f° is typically unknown. Let $f(x_1, \ldots, x_d; \boldsymbol{\eta})$ be our parametric approximation of f°, where $\boldsymbol{\eta} = (\boldsymbol{\theta}, \boldsymbol{\gamma})$ denotes the full parameter vector of the model, encompassing both the dependence parameters $\boldsymbol{\theta}$ and the marginal parameters $\boldsymbol{\gamma} = (\boldsymbol{\gamma}_1, \ldots, \boldsymbol{\gamma}_d)$. Using Equation (1), the joint log-likelihood function for the sample is given by:

$$\ell(\boldsymbol{\eta}) = \sum_{i=1}^{n} \log c\left(F_1(X_{i,1}; \boldsymbol{\gamma}_1), \ldots, F_d(X_{i,d}; \boldsymbol{\gamma}_d); \boldsymbol{\theta}\right) + \sum_{i=1}^{n} \sum_{l=1}^{d} \log f_l(X_{i,l}; \boldsymbol{\gamma}_l).$$

When parametric families for the copula and the margins are chosen, the two-stage MLE works as follows.

1. **Stage 1 (Marginal Estimation):** For each $l = 1, \ldots, d$, obtain the marginal maximum likelihood estimate $\widetilde{\boldsymbol{\gamma}}_l$ by maximizing the marginal log-likelihood with respect to $\boldsymbol{\gamma}_l$:

$$\widetilde{\boldsymbol{\gamma}}_l = \arg\max_{\boldsymbol{\gamma}_l} \sum_{i=1}^{n} \log f_l(X_{i,l}; \boldsymbol{\gamma}_l).$$

2. **Stage 2 (Copula Estimation):** Plug in $\widetilde{\boldsymbol{\gamma}} = (\widetilde{\boldsymbol{\gamma}}_1, \ldots, \widetilde{\boldsymbol{\gamma}}_d)$ from Stage 1, to get

$$\ell(\boldsymbol{\theta}, \widetilde{\boldsymbol{\gamma}}) = \sum_{i=1}^{n} \log c\left(F_1\left(X_{i,1}; \widetilde{\boldsymbol{\gamma}}_1\right), \ldots, F_d\left(X_{i,d}; \widetilde{\boldsymbol{\gamma}}_d\right); \boldsymbol{\theta}\right) + \sum_{i=1}^{n} \sum_{l=1}^{d} \log f_l(X_{i,l}; \widetilde{\boldsymbol{\gamma}}_l).$$

   Then, $\boldsymbol{\theta}$ is estimated by maximizing $\ell(\boldsymbol{\theta}, \widetilde{\boldsymbol{\gamma}})$ with respect to $\boldsymbol{\theta}$, yielding

$$\widetilde{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, \widetilde{\boldsymbol{\gamma}}).$$

The final two-stage maximum likelihood estimate for the entire model is

$$\widetilde{\boldsymbol{\eta}} = (\widetilde{\boldsymbol{\theta}}, \widetilde{\gamma}).$$

2.2 Two-Stage MLE Information Criteria: $\mathrm{CIC}_{\mathbf{2MLE}}$ and $\mathrm{AIC}_{\mathbf{2MLE}}$

Consider $B \in \mathbb{N}$ parametric families

$$\mathcal{F}_b = \left\{ f(\cdot; \boldsymbol{\theta}_b, \gamma_{b,1}, \ldots, \gamma_{b,d}) : (\boldsymbol{\theta}_b, \gamma_{b,1}, \ldots, \gamma_{b,d}) \in \Theta_b \times \Gamma_{b,1} \ldots \times \Gamma_{b,d} \right\} \quad \text{for } b = 1, \ldots, B,$$

for a given dataset $\{\boldsymbol{X}_i\}_{i=1}^n$, where the joint parameters $\boldsymbol{\eta}_b = (\boldsymbol{\theta}_b, \gamma_{b,1}, \ldots, \gamma_{b,d})$ are estimated via classical maximum likelihood estimation. The resulting maximum likelihood estimates for family b are denoted by $\widehat{\boldsymbol{\eta}}_b$. Note that we use a hat $\widehat{}$ to indicate classical MLEs and a tilde $\widetilde{}$ to indicate two-stage MLEs. To select the best fitted model, one can evaluate the $\mathrm{TIC}_{\mathrm{MLE}}$:

$$\mathrm{TIC}_{\mathrm{MLE}}(b) = 2\ell(\widehat{\boldsymbol{\eta}}_b) - 2\widehat{p}_{\boldsymbol{\eta}_b},$$

where $\widehat{p}_{\boldsymbol{\eta}_b} = \mathrm{tr}\left(\widehat{I}_{\boldsymbol{\eta}_b}^{-1} \widehat{K}_{\boldsymbol{\eta}_b}\right)$ is a bias-correction term, $\widehat{I}_{\boldsymbol{\eta}_b}$ is the empirical Fisher information matrix, and $\widehat{K}_{\boldsymbol{\eta}_b}$ is the estimated covariance matrix of the score vector. If the true data-generating density f° belongs to the parametric family $\mathcal{F}_b$, then the $\mathrm{TIC}_{\mathrm{MLE}}$ simplifies to the $\mathrm{AIC}_{\mathrm{MLE}}$:

$$\mathrm{AIC}_{\mathrm{MLE}}(b) = 2\ell(\widehat{\boldsymbol{\eta}}_b) - 2\mathrm{dim}(\boldsymbol{\eta}_b). \tag{2}$$

Using the information criterion $\mathrm{TIC}_{\mathrm{MLE}}$ is theoretically valid for ranking the fitted models. However, estimating all parameters simultaneously using full MLE $\widehat{\boldsymbol{\eta}}$ is often computationally expensive for high-dimensional data. Therefore, a common alternative approach is to use the two-stage MLE $\widetilde{\boldsymbol{\eta}}$. If one wishes to rank the fitted models using these two-stage estimates $\widetilde{\boldsymbol{\eta}}$, the standard $\mathrm{TIC}_{\mathrm{MLE}}$ formula cannot be applied, because the classical bias-correction term $\widehat{p}_{\boldsymbol{\eta}}$ is no longer theoretically valid under two-stage MLE. In Ko and Hjort (2019), the copula information criterion $\mathrm{CIC}_{\mathrm{2MLE}}$ was derived for two-stage MLE. This criterion is an analogue of the $\mathrm{TIC}_{\mathrm{MLE}}$ in the sense that it does not assume that $\mathcal{F}_b$ includes f°:

$$\mathrm{CIC}_{\mathrm{2MLE}}(b) = 2\ell(\widetilde{\boldsymbol{\eta}}_b) - 2\widetilde{p}_{\boldsymbol{\eta}_b},$$

where $\widetilde{p}_{\boldsymbol{\eta}_b}$ is a bias-correction term whose complex analytical representation will be provided in the next subsection. The authors also showed that if $\mathcal{F}_b$ includes f°, the bias-correction term $\widetilde{p}_{\boldsymbol{\eta}_b}$ does not need to be computed. The $\mathrm{CIC}_{\mathrm{2MLE}}$ simply reduces to:

$$\mathrm{AIC}_{\mathrm{2MLE}}(b) = 2\ell(\widetilde{\boldsymbol{\eta}}_b) - 2\mathrm{dim}(\boldsymbol{\eta}_b).$$

This means that the naive adaptation of $\mathrm{AIC}_{\mathrm{MLE}}$ to the two-stage MLE is theoretically justified, assuming the correct specification of the parametric family.

2.3 The Bias Correction Term of the $\mathrm{CIC}_{\mathbf{2MLE}}$

The Copula Information Criterion $\mathrm{CIC}_{\mathrm{2MLE}}$ relies on a model-robust bias correction term, denoted as $\widetilde{p}_{\boldsymbol{\eta}}$. This total penalty is decomposed into the sum of individual marginal penalties and a joint copula penalty. Analytically, it is expressed as:

$$\widetilde{p}_{\boldsymbol{\eta}} = \sum_{l=1}^{d} \widetilde{p}_{\gamma_l} + \widetilde{p}_{\boldsymbol{\theta}},$$

where $\widetilde{p}_{\gamma_l}$ is the penalty for the l-th marginal distribution, and $\widetilde{p}_{\boldsymbol{\theta}}$ is the penalty associated with the copula parameters. Here, we present only the empirical formulations of the penalties. The theoretical penalties are the probability limits of these empirical counterparts. In other words, the empirical penalties are consistent estimators of the theoretical penalties, obtained by computing plug-in sample averages evaluated at the two-stage ML parameter estimates $\widetilde{\boldsymbol{\eta}} = \left(\widetilde{\boldsymbol{\theta}}, \widetilde{\gamma}_1, \dots, \widetilde{\gamma}_d\right)$. Given a sample of n independent observations $\boldsymbol{X}_i = (X_{i,1}, \dots, X_{i,d})^T$ for $i = 1, \dots, n$, the empirical formulations are derived as follows.

2.3.1 Empirical Marginal Penalties $\widetilde{p}_{\gamma_l}$

For each margin l, the penalty is given by the trace of the product of the inverse empirical Fisher information matrix and the empirical score covariance matrix:

$$\widetilde{p}_{\gamma_l} = \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\gamma_l}^{-1} \widetilde{K}_{\gamma_l}\right)$$

The components of the above equation are evaluated using the empirical marginal score $U_{\gamma_l}(X_{i,l}; \widetilde{\gamma}_l) = \frac{\partial \log f_l(X_{i,l};\widetilde{\gamma}_l)}{\partial \gamma_l}$ and the marginal Hessian $H_{\gamma_l}(X_{i,l}; \widetilde{\gamma}_l) = \frac{\partial^2 \log f_l(X_{i,l};\widetilde{\gamma}_l)}{\partial \gamma_l \partial \gamma_l^T}$:

- Empirical Score Covariance:

$$\widetilde{K}_{\gamma_l} = \frac{1}{n} \sum_{i=1}^{n} U_{\gamma_l}(X_{i,l}; \widetilde{\gamma}_l) U_{\gamma_l}(X_{i,l}; \widetilde{\gamma}_l)^T.$$

- Empirical Information Matrix:

$$\widetilde{\mathcal{I}}_{\gamma_l} = -\frac{1}{n} \sum_{i=1}^{n} H_{\gamma_l}(X_{i,l}; \widetilde{\gamma}_l).$$

2.3.2 Empirical Copula Penalty $\widetilde{p}_{\boldsymbol{\theta}}$

Let us denote the copula log-density for the i-th observation evaluated at the two-stage MLEs $\widetilde{\boldsymbol{\eta}} = (\widetilde{\boldsymbol{\theta}}, \widetilde{\gamma})$ as follows:

$$\ell_i^{(c)} = \log c\left(F_1\left(X_{i,1}; \widetilde{\gamma}_1\right), \dots, F_d\left(X_{i,d}; \widetilde{\gamma}_d\right); \widetilde{\boldsymbol{\theta}}\right).$$

The following are the copula scores with respect to the marginal and copula parameters, respectively:

$$U_{\gamma}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}}) = \frac{\partial \ell_i^{(c)}}{\partial \gamma} \quad \text{and} \quad U_{\boldsymbol{\theta}}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}}) = \frac{\partial \ell_i^{(c)}}{\partial \boldsymbol{\theta}}.$$

Furthermore, let us aggregate all the marginal scores into a single large score vector for observation i:

$$U_{\gamma}(\boldsymbol{X}_i; \widetilde{\gamma}) = (U_{\gamma_1}(X_{i,1}; \widetilde{\gamma}_1)^T, \dots, U_{\gamma_d}(X_{i,d}; \widetilde{\gamma}_d)^T)^T$$

To establish the empirical copula penalty $\widetilde{p}_{\boldsymbol{\theta}}$, we will need the following empirical matrices.

- Cross-Score Covariances:

$$\widetilde{K}_{\gamma}^{\circ} = \frac{1}{n} \sum_{i=1}^{n} U_{\gamma}(\boldsymbol{X}_i; \widetilde{\gamma}) U_{\gamma}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}})^T,$$

$$\widetilde{K}_{\gamma,\boldsymbol{\theta}} = \frac{1}{n} \sum_{i=1}^{n} U_{\gamma}(\boldsymbol{X}_i; \widetilde{\gamma}) U_{\boldsymbol{\theta}}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}})^T,$$

$$\widetilde{K}_{\boldsymbol{\theta}} = \frac{1}{n} \sum_{i=1}^{n} U_{\boldsymbol{\theta}}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}}) U_{\boldsymbol{\theta}}^{(c)}(\boldsymbol{X}_i; \widetilde{\boldsymbol{\eta}})^T.$$

- Copula Information Sub-Matrices:

$$\widetilde{\mathcal{I}}_{\boldsymbol{\gamma},\boldsymbol{\theta}} = -\frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2 \ell_i^{(c)}}{\partial\boldsymbol{\gamma}\partial\boldsymbol{\theta}^T},$$

$$\widetilde{\mathcal{I}}_{\boldsymbol{\theta}} = -\frac{1}{n}\sum_{i=1}^{n}\frac{\partial^2 \ell_i^{(c)}}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}.$$

Additionally, denote $\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}}$ as a block-diagonal matrix combined by the previously computed marginal information matrices:

$$\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}} = \begin{pmatrix} \widetilde{\mathcal{I}}_{\gamma_1} & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \widetilde{\mathcal{I}}_{\gamma_2} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \widetilde{\mathcal{I}}_{\gamma_d} \end{pmatrix}.$$

Based on the calculations provided on page 172 of Ko and Hjort (2019), the empirical copula penalty is given by:

$$\widetilde{p}_{\boldsymbol{\theta}} = \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}}^{-1}\widetilde{K}_{\boldsymbol{\gamma}}^{\circ}\right) - \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\theta}}^{-1}\widetilde{\mathcal{I}}_{\boldsymbol{\gamma},\boldsymbol{\theta}}^{T}\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}}^{-1}\widetilde{K}_{\boldsymbol{\gamma},\boldsymbol{\theta}}\right) + \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\theta}}^{-1}\widetilde{K}_{\boldsymbol{\theta}}\right).$$

Thus, the overall bias-correction term for the $\mathrm{CIC}_{2\mathrm{MLE}}$ is given by

$$\widetilde{p}_{\boldsymbol{\eta}} = \sum_{l=1}^{d}\operatorname{tr}\left(\widetilde{\mathcal{I}}_{\gamma_l}^{-1}\widetilde{K}_{\gamma_l}\right) + \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}}^{-1}\widetilde{K}_{\boldsymbol{\gamma}}^{\circ}\right) - \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\theta}}^{-1}\widetilde{\mathcal{I}}_{\boldsymbol{\gamma},\boldsymbol{\theta}}^{T}\widetilde{\mathcal{I}}_{\boldsymbol{\gamma}}^{-1}\widetilde{K}_{\boldsymbol{\gamma},\boldsymbol{\theta}}\right) + \operatorname{tr}\left(\widetilde{\mathcal{I}}_{\boldsymbol{\theta}}^{-1}\widetilde{K}_{\boldsymbol{\theta}}\right).$$

2.4 Simulation Study: Performance of $\mathrm{CIC}_{2\mathrm{MLE}}$ and $\mathrm{AIC}_{2\mathrm{MLE}}$

To investigate the behavior of the $\mathrm{CIC}_{2\mathrm{MLE}}$ and the $\mathrm{AIC}_{2\mathrm{MLE}}$, we independently replicate the first simulation study presented in Ko and Hjort (2019). The primary objective of this simulation is to examine the similarity between the $\mathrm{CIC}_{2\mathrm{MLE}}$ and the $\mathrm{AIC}_{2\mathrm{MLE}}$, and to determine whether models with a higher $\mathrm{CIC}_{2\mathrm{MLE}}$ or $\mathrm{AIC}_{2\mathrm{MLE}}$ value are indeed "better" models in practice. Then the behavior of the bias-correction term $\widetilde{p}_{\boldsymbol{\eta}}$ is compared to the naive penalty $\dim(\boldsymbol{\eta})$.

2.4.1 Simulation Setup and Candidate Models

We assume that the true data-generating density f° is a 4-dimensional pdf ($d = 4$). The true dependence structure is governed by a Gumbel copula with parameter $\theta^{\circ} = 3$, which exhibits strong upper-tail dependence. The true marginal distributions are configured as follows:

- Margin 1: Weibull with $\gamma_1^{\circ} = 1.5$ (shape) and $\gamma_2^{\circ} = 4$ (scale).
- Margin 2: Weibull with $\gamma_3^{\circ} = 2$ (shape) and $\gamma_4^{\circ} = 3$ (scale).
- Margin 3: Gamma with $\gamma_5^{\circ} = 2$ (shape) and $\gamma_6^{\circ} = 1$ (rate).
- Margin 4: Gamma with $\gamma_7^{\circ} = 3$ (shape) and $\gamma_8^{\circ} = 1$ (rate).

From this true model, we generate $n = 1000$ independent observations for each simulation run. To evaluate the model selection criteria, we define a candidate set of models consisting of $B = 324$ distinct fully parametric models, which are based on all possible combinations of the candidate copulas and margins described in Table 1.

	Candidates
Copula	Gumbel, Gaussian, Frank, Survival Clayton
Margin 1	Weibull, Gamma, Log-normal
Margin 2	Weibull, Gamma, Normal
Margin 3	Gamma, Weibull, Log-normal
Margin 4	Gamma, Weibull, Log-normal

Table 1: List of the candidate copulas and margins.

2.4.2 Evaluation Metric: The Joint Tail Probability

Note that $\mathrm{CIC}_{\mathrm{2MLE}}$ evaluates models based on their estimated Kullback-Leibler divergence from the true distribution, which is a purely theoretical concept. A natural question arises: does this criterion actually identify models that are useful in practice? To answer this, we first need to establish a metric that reflects practical usage. In fields like finance, insurance, and hydrology, modeling multivariate extremes (e.g., simultaneous market crashes or concurrent floods) is often the primary goal of copula modeling. The probability of simultaneous extreme events in the upper tail is highly sensitive to the choice of the copula. Therefore, accurate estimation of joint tail probabilities can serve as a relevant metric for evaluating a model's practical utility. Specifically, suppose we are interested in estimating the probability that all four variables simultaneously exceed their respective 0.7-quantiles:

$$\mathrm{P}\left(X_1 > q_{0.7}^{(1)}, X_2 > q_{0.7}^{(2)}, X_3 > q_{0.7}^{(3)}, X_4 > q_{0.7}^{(4)}\right), \tag{3}$$

where $q_{0.7}^{(l)} = (F_l^{\circ})^{-1}(0.7)$ represents the true 0.7-quantile of the l-th margin defined by the data-generating model. For each fitted model $b = 1, \ldots, 324$, we compute the estimated joint tail probability using two-stage MLE and record its Mean Squared Error (MSE) relative to the true probability across 1000 Monte-Carlo replications.

2.4.3 Computational Implementation Details

As outlined in the Section 2.3, computing the $\mathrm{CIC}_{\mathrm{2MLE}}$ requires evaluating the trace of several complex matrix products. In our implementation, to ensure numerical stability and computational speed, the marginal penalties $\widetilde{p}_{\boldsymbol{\gamma}_l}$ are calculated using exact analytical gradients (scores) and analytical Hessians for the Weibull, Gamma, Normal, and Log-normal distributions. However, for the joint copula penalty $\widetilde{p}_{\boldsymbol{\theta}}$, calculating analytical cross-derivatives between the copula parameter and the marginal parameters is highly intractable. Therefore, following standard practices, cross-score covariances and copula information sub-matrices ($\widetilde{K}_{\boldsymbol{\gamma}}^{\circ}$, $\widetilde{K}_{\boldsymbol{\gamma},\boldsymbol{\theta}}$, $\widetilde{K}_{\boldsymbol{\theta}}$, $\widetilde{\mathcal{I}}_{\boldsymbol{\gamma},\boldsymbol{\theta}}$, and $\widetilde{\mathcal{I}}_{\boldsymbol{\theta}}$) are approximated numerically using the `numDeriv` package in R. The simulation process was parallelized using the `foreach` and `doParallel` packages to handle the extensive computational load of evaluating 324 models across 1000 MC replications.

2.4.4 Results and Discussion: Predictive Performance under Two-Stage MLE

The results of our simulation study replicate the findings from Ko and Hjort (2019), confirming the behavior of both criteria $\mathrm{CIC}_{2\mathrm{MLE}}/\mathrm{AIC}_{2\mathrm{MLE}}$. Figure 1 plots the MSE of the estimated joint tail probability against the average values of the $\mathrm{CIC}_{2\mathrm{MLE}}$ and the $\mathrm{AIC}_{2\mathrm{MLE}}$ for all 324 candidate models across 1000 Monte Carlo replications. The results display a clear negative trend: models that achieve better (higher) values in both $\mathrm{CIC}_{2\mathrm{MLE}}$ and $\mathrm{AIC}_{2\mathrm{MLE}}$ yield lower MSEs. This confirms that minimizing the KL-divergence successfully identifies models with better estimation of the joint tail probability. Furthermore, we observe that the overall difference between $\mathrm{CIC}_{2\mathrm{MLE}}$ and $\mathrm{AIC}_{2\mathrm{MLE}}$ is remarkably minimal. This high degree of similarity occurs because, for the considered sample size of $n = 1000$, the shared log-likelihood term, $\ell(\widetilde{\boldsymbol{\eta}})$, is substantially larger in absolute value than the bias correction terms, $\widetilde{p}_{\boldsymbol{\eta}}$ and $\dim(\boldsymbol{\eta})$, and therefore dominates the criteria. The difference between $\mathrm{CIC}_{2\mathrm{MLE}}$ and $\mathrm{AIC}_{2\mathrm{MLE}}$ (the misalignment between the black triangles and the red crosses in Figure 1) is caused by the difference in their bias correction part alone.

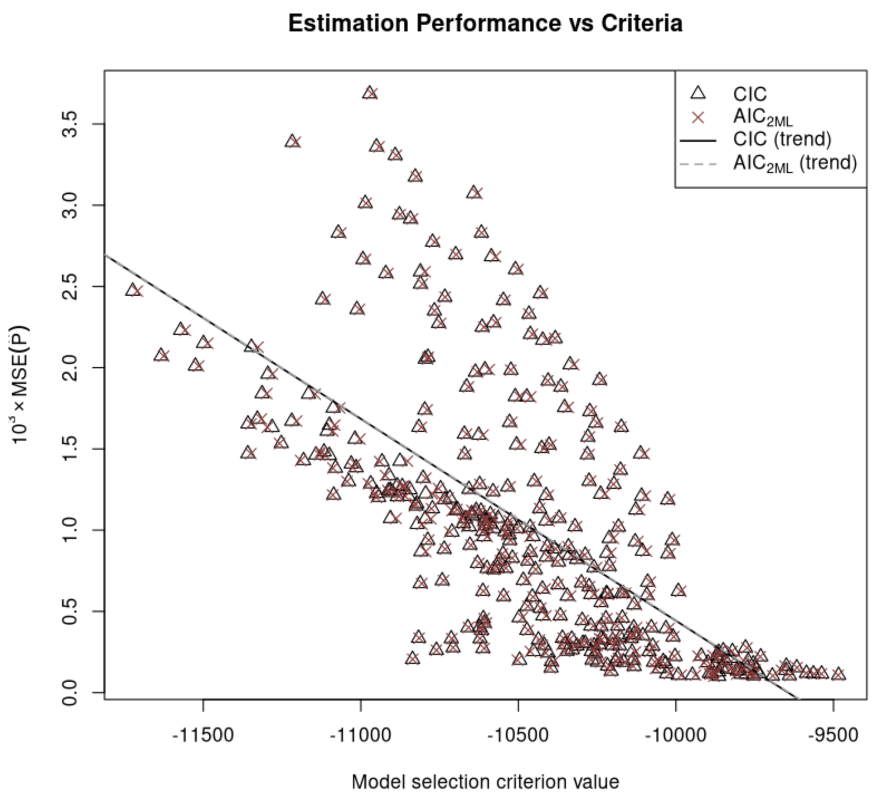

Figure 1: The Mean Squared Error of the estimated joint tail probability (3) versus the average values of the $\mathrm{CIC}_{2\mathrm{MLE}}$ and the $\mathrm{AIC}_{2\mathrm{MLE}}$ for all 324 candidate models across 1000 Monte Carlo replications. The parameters are estimated using two-stage maximum likelihood estimation. Note that the MSE values on the y-axis are scaled by a factor of 1000.

To isolate the differences between the two criteria, Figure 2 plots the MSE of the estimated joint tail probability against the average bias-correction terms, namely $\widetilde{p}_{\boldsymbol{\eta}}$ for the $\mathrm{CIC}_{2\mathrm{MLE}}$ and $\dim(\boldsymbol{\eta})$ for the $\mathrm{AIC}_{2\mathrm{MLE}}$, for all 324 candidate models across 1000 Monte Carlo replications. In this plot, the conceptual advantage of $\mathrm{CIC}_{2\mathrm{MLE}}$ becomes highly visible. For $\mathrm{AIC}_{2\mathrm{MLE}}$, the penalty term $\dim(\boldsymbol{\eta})$ is strictly a constant vertical line representing the parameter count, which is 9 in our setting. It possesses no dynamic relationship with the estimation accuracy of the model. Conversely, the robust penalty $\widetilde{p}_{\boldsymbol{\eta}}$ of $\mathrm{CIC}_{\mathrm{2MLE}}$ exhibits a strong positive correlation with the MSE. As a fitted model's MSE increases, the trace components of the empirical score covariance and information matrices (detailed in Section 2.3) grow, causing the $\mathrm{CIC}_{\mathrm{2MLE}}$ bias-correction term to penalize such models more heavily.

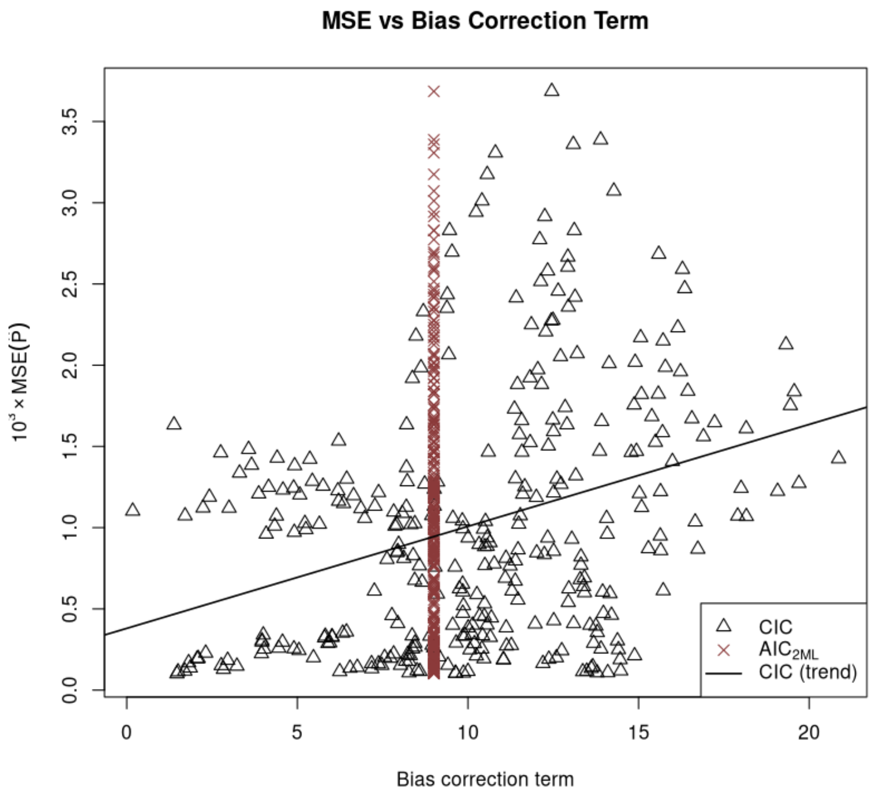

Figure 2: Mean squared error (MSE) of the estimated joint tail probability in Equation (3) plotted against the average bias-correction terms: $\widetilde{p}_{\boldsymbol{\eta}}$ for the $\mathrm{CIC}_{\mathrm{2MLE}}$ and $\dim(\boldsymbol{\eta})$ for the $\mathrm{AIC}_{\mathrm{2MLE}}$. The MSE and average penalties are computed across 1000 Monte Carlo replications, with parameters estimated via two-stage maximum likelihood. Note that the MSE values on the y-axis are scaled by a factor of 1000.

Overall, the simulation results indicate that the rankings produced by the $\mathrm{AIC}_{\mathrm{2MLE}}$ and the $\mathrm{CIC}_{\mathrm{2MLE}}$ coincide to a high degree. Consequently, the extra computational effort required to empirically estimate the correction term for the $\mathrm{CIC}_{\mathrm{2MLE}}$ yields little practical benefit. It is important to note, however, that the $\mathrm{CIC}_{\mathrm{2MLE}}$ is an asymptotic tool derived via a Taylor expansion; therefore, its theoretical validity relies heavily on the sample size n. Because our simulation used a large sample size ($n = 1000$), the two criteria naturally performed almost identically. For smaller sample sizes, the empirically estimated bias-correction term in the $\mathrm{CIC}_{\mathrm{2MLE}}$ could potentially dominate the likelihood term, leading to increased variability in the criterion's performance. For this reason, practitioners generally prefer AIC-type criteria.

3 Cross-Validation Strategies for Pseudo-Likelihood Estimation

In contrast to the fully parametric approach, the semiparametric framework avoids making parametric assumptions about the marginal distributions. However, a parametric assumption must still be made regarding the copula. This combination of nonparametric and parametric estimation explains the term 'semiparametric'. Such a strategy avoids the risk of misspecified marginals, which can severely bias the estimated copula parameters.

3.1 Maximum Pseudo-Likelihood Estimation (MPLE)

As in the previous section, let $\boldsymbol{X} = (X_1, \ldots, X_d)^T$ be a d-variate random vector drawn from an absolutely continuous distribution with cdf F°. By Sklar's Theorem, F° can be expressed as

$$F^\circ(\boldsymbol{x}) = C^\circ(F_1^\circ(x_1), \ldots, F_d^\circ(x_d)).$$

Assuming the relevant derivatives exist, the corresponding true pdf f° is given by

$$f^\circ(\boldsymbol{x}) = c^\circ(F_1^\circ(x_1), \ldots, F_d^\circ(x_d)) \prod_{l=1}^{d} f_l^\circ(x_l).$$

Suppose $\boldsymbol{X}_i = (X_{i,1}, \ldots, X_{i,d})^T$, for $i = 1, \ldots, n$, represent independent and identically distributed (i.i.d.) realizations drawn from the underlying distribution. Our objective is to fit a parametric model to the copula while leaving the marginal distributions unspecified. Let $\{c(\cdot; \boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$ be a candidate parametric copula family, our primary objective is to find a model that best approximates the true copula density c°. In the semiparametric setting, estimation begins by transforming the independent observations $\mathcal{X}_n = \{\boldsymbol{X}_i\}_{i=1}^n$ into dependent pseudo-observations, $\{\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i)\}_{i=1}^n \subset [0,1]^d$. Here, $\boldsymbol{F}_{n,\perp}$ denotes the vector of rescaled marginal empirical distribution functions:

$$\boldsymbol{F}_{n,\perp}(\boldsymbol{x}) = (F_{n,1}(x_1), \ldots, F_{n,d}(x_d)), \text{ where } F_{n,l}(x_l) = \frac{1}{n+1} \sum_{i=1}^{n} I(X_{i,l} \leq x_l).$$

The rescaling factor $1/(n+1)$, rather than the standard $1/n$, is employed to prevent evaluating $\boldsymbol{u} \mapsto \log c(\boldsymbol{u}; \boldsymbol{\theta})$ at the boundary of the unit hypercube $[0,1]^d$, where the densities of most copulas of interest diverge to infinity. Using these pseudo-observations, we construct the pseudo-log-likelihood function:

$${}^p\ell(\boldsymbol{\theta}) = \sum_{i=1}^{n} \log c(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i); \boldsymbol{\theta}).$$

The standard estimator for the copula parameter, known as the maximum pseudo-likelihood estimator (MPLE), is then defined as the maximizer of this pseudo-log-likelihood

$${}^p\widehat{\boldsymbol{\theta}} = \arg\max_{\boldsymbol{\theta} \in \Theta} {}^p\ell(\boldsymbol{\theta}).$$

Under suitable regularity conditions, such as those established by Genest et al. (1995), this maximizer exhibits the same asymptotic behavior as the classical MLE. Specifically, the following convergence in probability holds:

$${}^p\widehat{\boldsymbol{\theta}} \xrightarrow[n\to\infty]{\mathrm{P}} \arg\min_{\boldsymbol{\theta} \in \Theta} \int_{[0,1]^d} \log \left[\frac{c^\circ(\boldsymbol{u})}{c(\boldsymbol{u}; \boldsymbol{\theta})} \right] dC^\circ(\boldsymbol{u}),$$

where the integral on the right-hand side represents the Kullback-Leibler (KL) divergence between the true copula density c° and the candidate model $c(\cdot; \boldsymbol{\theta})$.

3.2 Semiparametric Information Criteria

Consider $B \in \mathbb{N}$ candidate parametric copula families:

$$\mathcal{C}_b = \{c(\cdot; \boldsymbol{\theta}_b) : \boldsymbol{\theta}_b \in \Theta_b\} \quad \text{for } b = 1, \ldots, B.$$

For a given dataset $\mathcal{X}_n$, we estimate the dependence parameters using MPLE, and our goal is to rank the fitted models. Note that model selection is more complex here than in the fully parametric framework. By transforming the original data into pseudo-observations, we add another level of uncertainty, and a correct model selection procedure must account for it. As indicated in the introduction, practitioners working in the semiparametric framework commonly rely on a naive adaptation of the Akaike Information Criterion Akaike (1974), which is given by:

$$^{p}\mathrm{AIC} = 2\left[{}^{p}\ell\left({}^{p}\widehat{\boldsymbol{\theta}}\right) - \dim(\boldsymbol{\theta})\right]. \tag{4}$$

This criterion is purely heuristic and lacks theoretical justification, as it ignores the semiparametric nature of the problem. The initial motivation for this criterion was that as $n \to \infty$, $\boldsymbol{F}_{n,\perp}(\boldsymbol{x})$ converges to $(F_1^{\circ}(x_1), \ldots, F_d^{\circ}(x_d))$ for all $\boldsymbol{x} = (x_1, \ldots, x_d) \in \mathbb{R}^d$. Consequently, it was initially expected that minor modifications to parametric copula model selection techniques, such as Equation (2), should still provide a decent ranking of the candidate models. However, it was shown in Grønneberg and Hjort (2014) that ^{p}AIC is theoretically invalid for most common copula families. In fact, the authors proved a much stronger result: a generally applicable AIC-like model selection formula does not even exist for many popular copula models under MPLE. Specifically, when deriving the exact bias-correction term required for the MPLE, they discovered that the random variable responsible for the systematic deviation lacks a finite first moment for most popular copula models. This means that in the copula context, when viewing model selection from a loss function perspective, standard penalty-based criteria are simply not suited for pseudo-likelihood estimation. Nevertheless, the formula (4) remains widely used due to its computational simplicity.

3.2.1 Leave-One-Out Cross-Validation $^{p}\mathrm{xv}_1$

It is well known that the $\mathrm{TIC}_{\mathrm{MLE}}$ formula is first-order equivalent to leave-one-out cross-validation [see Chapter 2 of Claeskens and Hjort (2008)]:

$$\mathrm{TIC}_{\mathrm{MLE}} = 2n \cdot \mathrm{xv}_1 + o_P(1), \tag{5}$$

where $\mathrm{xv}_1 = \frac{1}{n}\sum_{i=1}^{n} \log\left[f\left(\boldsymbol{X}_i; \widehat{\boldsymbol{\eta}}_{(-i)}\right)\right]$, and $\widehat{\boldsymbol{\eta}}_{(-i)}$ is the MLE based on the sample $\mathcal{X}_n$ without the i-th observation $\boldsymbol{X}_i$. However, in Section 4 of Grønneberg and Hjort (2014), it was shown that in the case of the MPLE, an equivalence analogous to (5) does not hold, which provides a further contrast between MPLE-based and MLE-based estimation. The authors used this non-equivalence as an alternative approach to copula model selection. They used the leave-one-out cross-validation formula as a model selection criterion from a prediction perspective, which is applicable to any copula model. The exact leave-one-out cross-validation criterion is defined as:

$$^{p}\mathrm{xv}_1 = \frac{1}{n}\sum_{i=1}^{n} \log\left[c\left(\boldsymbol{F}_{n,\perp,(-i)}(\boldsymbol{X}_i); {}^{p}\widehat{\boldsymbol{\theta}}_{(-i)}\right)\right], \text{ where} \tag{6}$$

- $\boldsymbol{F}_{n,\perp,(-i)}(\boldsymbol{X}_i) = \left(F_{(-i),1}(X_{i,1}), \ldots, F_{(-i),d}(X_{i,d})\right)$, where $F_{(-i),l}$ is the $\frac{n-1}{n}$-rescaled ecdf of the l-th marginal, computed from the sample $\mathcal{X}_n$ excluding $\boldsymbol{X}_i$, for $l = 1, \ldots, d$,

- ${}^{p}\widehat{\boldsymbol{\theta}}_{(-i)} = \underset{\boldsymbol{\theta}\in\Theta}{\text{argmax}} \ \sum_{j\neq i} \log\left[c\left(\boldsymbol{F}_{n,\perp,(-i)}(\boldsymbol{X}_j);\boldsymbol{\theta}\right)\right]$.

Note that calculating ${}^{p}\text{xv}_1$ requires refitting the MPLE n separate times, which is computationally expensive for large n. Furthermore, if $\dim(\boldsymbol{\theta})$ is large, the ${}^{p}\text{xv}_1$ would require solving numerous, potentially challenging numerical optimizations. If some of these optimizations do not identify the correct maximum pseudo-likelihood solution, this may skew the entire computation of the ${}^{p}\text{xv}_1$.

3.2.2 Analytical Representation of xv_{CIC}

Because exact calculation of ${}^{p}\text{xv}_1$ is computationally very expensive, Grønneberg and Hjort (2014) derived its first-order asymptotic approximation, denoted as xv_{CIC}. This criterion is formally given by:

$$\text{xv}_{\text{CIC}} = 2\,{}^{p}\ell\left({}^{p}\widehat{\boldsymbol{\theta}}\right) - 2\left(\widehat{\delta}_c + \widehat{\delta}_m\right),$$

where $\widehat{\delta}_c$ and $\widehat{\delta}_m$ are bias-correction terms, whose explicit formulas will be provided later. Note that the xv_{CIC} formula was derived via an asymptotic approximation of the cross-validation formula (6); hence, it is only valid for sufficiently large n. Strictly speaking, $\text{xv}_{\text{CIC}}/2$ is first-order approximation to the leave-one-out cross-validation criterion ${}^{p}\text{xv}_1$, where the multiplication by 2 is included to maintain the traditional scale of the Akaike Information Criterion. Similar to ${}^{p}\text{xv}_1$, the xv_{CIC} criterion applies to all common copula models. Furthermore, while the xv_{CIC} formula features the familiar penalized-likelihood structure, it is motivated purely from a prediction perspective rather than a Kullback-Leibler divergence perspective.

Before explicitly stating the formulas for the bias-correction terms $\widehat{\delta}_c$ and $\widehat{\delta}_m$, we must define the following preliminary quantities:

- $\phi(\boldsymbol{u};\boldsymbol{\theta}) = \frac{\partial \log c(\boldsymbol{u};\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$,
- $\zeta'(\boldsymbol{u};\boldsymbol{\theta}) = \frac{\partial \log c(\boldsymbol{u};\boldsymbol{\theta})}{\partial \boldsymbol{u}}$,
- $\widehat{J}_n = \left(-\frac{1}{n}\right)\sum_{i=1}^{n} \frac{\partial^2 \log c\left(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);{}^{p}\widehat{\boldsymbol{\theta}}\right)}{\partial\boldsymbol{\theta}\partial\boldsymbol{\theta}^T}$,
- $\widehat{z}(\boldsymbol{x};\boldsymbol{\theta}) = \frac{1}{n}\sum_{l=1}^{d}\sum_{i=1}^{n} \frac{\partial \phi(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);\boldsymbol{\theta})}{\partial u_l}\cdot\left[\mathbf{1}\{x_l \le F_{n,l}(X_{i,l})\} - F_{n,l}(X_{i,l})\right]$.

The correction terms are then calculated as:

$$\widehat{\delta}_c = \frac{1}{n}\sum_{i=1}^{n} \phi\left(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);{}^{p}\widehat{\boldsymbol{\theta}}\right)^T \widehat{J}_n^{-1}\left[\phi\left(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);{}^{p}\widehat{\boldsymbol{\theta}}\right) + \widehat{z}\left(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);{}^{p}\widehat{\boldsymbol{\theta}}\right)\right],$$

$$\widehat{\delta}_m = \frac{1}{n}\sum_{i=1}^{n} \zeta'\left(\boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i);{}^{p}\widehat{\boldsymbol{\theta}}\right)^T\left[\mathbf{1}_d - \boldsymbol{F}_{n,\perp}(\boldsymbol{X}_i)\right],$$

where $\mathbf{1}_d$ denotes a d dimensional vector of ones. One can see that computing the xv_{CIC} does not require refitting the MPLE n separate times, unlike the ${}^{p}\text{xv}_1$. Actually, it only requires single numerical optimization to obtain ${}^{p}\widehat{\boldsymbol{\theta}}$. However, the computation of the xv_{CIC} can still be quite demanding due to the bias-correction terms $\widehat{\delta}_c$ and $\widehat{\delta}_m$. Therefore, one might question the practical advantage of using the xv_{CIC} over the ${}^{p}\text{AIC}$. Jordanger and Tjøstheim (2014) explored this in a simulation study, where they observed only minor differences between the two. We aim to independently replicate their study to verify these findings. Additionally, we expand upon their work by incorporating the most computationally demanding criterion ${}^{p}\mathrm{xv}_1$. Although the ${}^{p}\mathrm{xv}_1$ is computationally prohibitive for practical applications, including it in our simulation can serve a valuable illustrative purpose.

3.2.3 Proposed Method: K_n-Fold Cross-Validation

In the previous section 3.2.2, the primary motivation for defining the $\mathrm{xv}_{\mathrm{CIC}}$ was to obtain a first-order asymptotic approximation of the ${}^{p}\mathrm{xv}_1$, given that the latter becomes computationally prohibitive for large sample sizes n. Previous simulations by Jordanger and Tjøstheim (2014) demonstrated that the $\mathrm{xv}_{\mathrm{CIC}}$ offers virtually no practical advantage over the theoretically invalid ${}^{p}\mathrm{AIC}$. For exploratory purposes in this project, we propose a heuristic approximation of the ${}^{p}\mathrm{xv}_1$ by utilizing a K_n-fold cross-validation procedure with data-driven choices for the number of folds K_n. We randomly partition the training data indices $\{1, \dots, n\}$ into K_n validation folds of roughly equal size $n_v = n/K_n$. Let $\mathcal{I}_k$ denote set of indices belonging to the k-th fold for $k = 1, \dots, K_n$. Then, the K_n-fold criterion is defined as

$$ {}^{p}\mathrm{xv}_{K_n} = \frac{1}{n} \sum_{k=1}^{K_n} \sum_{i \in \mathcal{I}_k} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(\boldsymbol{X}_i); {}^{p}\widehat{\boldsymbol{\theta}}_{(-\mathcal{I}_k)} \right) \right], \text{ where} \tag{7} $$

- ${}^{p}\widehat{\boldsymbol{\theta}}_{(-\mathcal{I}_k)} = \underset{\boldsymbol{\theta} \in \Theta}{\operatorname{argmax}} \sum_{j \notin \mathcal{I}_k} \log \left[c \left(\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(\boldsymbol{X}_j); \boldsymbol{\theta} \right) \right]$,
- $\boldsymbol{F}_{n,\perp,(-\mathcal{I}_k)}(\boldsymbol{X}_i) = \left(F_{(-\mathcal{I}_k),1}(X_{i,1}), \dots, F_{(-\mathcal{I}_k),d}(X_{i,d}) \right)$, where $F_{(-\mathcal{I}_k),l}$ is the $\frac{n-n_v}{n}$-rescaled ecdf of the l-th marginal, computed from the sample $\mathcal{X}_n$ excluding $\{\boldsymbol{X}_j : j \in \mathcal{I}_k\}$, for $l = 1, \dots, d$.

The primary motivation behind the criterion in (7) is that leave-one-out cross-validation ${}^{p}\mathrm{xv}_1$ represents a special case of this framework in which $K_n = n$, corresponding to a validation set size of $n_v = 1$. Consequently, the ${}^{p}\mathrm{xv}_1$ is the most computationally demanding formulation of K_n-fold cross-validation. Note that for the ${}^{p}\mathrm{xv}_1$, the ratio of the validation set size to the full sample size asymptotically approaches zero:

$$ \frac{n_v}{n} = \frac{1}{n} \underset{n \to \infty}{\longrightarrow} 0. $$

Standard K-fold cross-validation (e.g., $K = 5$ or $K = 10$) violates this property, as $n_v/n = 1/K$ remains strictly bounded away from zero. As discussed previously, the ${}^{p}\mathrm{xv}_1$ criterion is theoretically valid in the context of copulas and MPLE, as it can be applied to any candidate copula model. Thus, to approximate the ${}^{p}\mathrm{xv}_1$ while mitigating its prohibitive computational cost, we investigate data-driven choices for the number of folds K_n that satisfy the limiting behavior $n_v/n = 1/K_n \underset{n \to \infty}{\longrightarrow} 0$. Specifically, we benchmark the following dynamic choices:

$$ K_n \in \left\{ \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\}. $$

For comprehensive comparison, our simulations also evaluate fixed choices

$$ K \in \{2, 5, 10\}. $$

3.3 Simulation Study: Semiparametric Copula Model Selection

The main objective of this simulation study is to replicate the simulations conducted by Jordanger and Tjøstheim (2014) to confirm that the $\mathrm{xv}_{\mathrm{CIC}}$ is not superior to the ${}^{p}\mathrm{AIC}$. Furthermore, we evaluate additional criteria, such as the ${}^{p}\mathrm{xv}_1$ and the ${}^{p}\mathrm{xv}_{K_n}$, for different choices of the number of folds K_n. Our goal is to take a detailed look at all the aforementioned criteria to see how they behave across various true data-generating copula models, degrees of dependence, and sample sizes.

3.3.1 Data Generation and Candidate Models

We constrain our analysis to bivariate distributions ($d = 2$). We simulate true data-generating models from five widely used one-parameter copula families:

Clayton, Gumbel, Joe, Frank, Gaussian.

To capture a diverse spectrum of dependence structures, each copula family is parameterized corresponding to seven different levels of Kendall's tau:

$$\tau \in \{0.05, 0.10, 0.15, 0.20, 0.25, 0.50, 0.75\}.$$

These values of Kendall's tau represent magnitudes of dependence ranging from extremely weak (e.g., $\tau = 0.05$) to strong (e.g., $\tau = 0.75$). This yields a total of $5 \times 7 = 35$ distinct true data-generating configurations. For each true model configuration, we evaluate the performance of the selection criteria across three different sample sizes: $n \in \{100, 200, 400\}$. For each combination of true model, τ, and sample size n, we conduct 1000 independent Monte Carlo replications. During each replication, all five candidate copula families are fitted to the generated data using the MPLE procedure.

3.3.2 Evaluation and Computational Implementation

The criteria evaluated during each replication include ${}^{p}\mathrm{AIC}$, ${}^{p}\mathrm{xv}_1$, $\mathrm{xv}_{\mathrm{CIC}}$ and ${}^{p}\mathrm{xv}_{K_n}$ with the following choices of number of folds

$$K_n \in \left\{2, 5, 10, \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\}.$$

Because calculating the exact $\mathrm{xv}_{\mathrm{CIC}}$ penalty requires evaluating highly complex partial derivatives of copula log-densities (see Subsection 3.2.2), we used the computer algebra system `Yacas` (via the `Ryacas` R package) to derive explicit analytical formulas for all required symbolic derivatives of the candidate copula densities. After obtaining the correct formulas, all functions required to compute $\mathrm{xv}_{\mathrm{CIC}}$ were stored in the file "explicit_derivatives_xv_CIC.R". Also, due to the heavy computational demands of running ${}^{p}\mathrm{xv}_1$ and ${}^{p}\mathrm{xv}_{K_n}$ over $35 \times 1000 = 35000$ simulated datasets per sample size, the entire simulation infrastructure was parallelized using the `doParallel` framework in R.

3.3.3 Hit Rates

To evaluate and compare the practical performance of the established criteria ${}^{p}\mathrm{AIC}$, ${}^{p}\mathrm{xv}_1$, $\mathrm{xv}_{\mathrm{CIC}}$ and ${}^{p}\mathrm{xv}_{K_n}$, we use the "hit rate" as our primary evaluation metric. The hit rate is defined as the empirical frequency with which a given criterion successfully selects the true data-generating copula family from the candidate set. We define the finite set of candidate parametric copula families be denoted by

$$\mathcal{C}_{\text{cand}} = \{\text{Clayton, Gumbel, Joe, Frank, Gaussian}\}.$$

The total number of Monte Carlo replications is fixed at $M = 1000$ for each specific combination of true copula family $c^\circ \in \mathcal{C}_{\text{cand}}$, sample size n, and Kendall's tau τ. During the m-th replication ($m = 1, \dots, M$), every candidate family $c \in \mathcal{C}_{\text{cand}}$ is fitted to the generated data. We then consider a generic information criterion, crit, belonging to the set:

$$\text{crit} \in \left\{ {}^p\text{AIC},\ {}^p\text{xv}_1,\ \text{xv}_{\text{CIC}},\ {}^p\text{xv}_{K_n} : K_n \in \left\{ 2, 5, 10, \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor, \lfloor \sqrt{n} \rfloor, \left\lfloor \frac{n}{\log(n)} \right\rfloor \right\} \right\}.$$

Let $S_m(\text{crit}, c)$ represent the numerical score assigned by that criterion to the fitted candidate family c. Note that the criteria are formulated such that higher scores indicate a better fit. Therefore, the specific copula model chosen by a given criterion in the m-th replication, denoted as $c^{\text{selected}}_{m,\text{crit}}$, is formally defined as the family that maximizes the criterion's score:

$$c^{\text{selected}}_{m,\text{crit}} = \underset{c \in \mathcal{C}\text{cand}}{\arg\max}\ S_m(\text{crit}, c).$$

Once the optimal candidate family is selected by the criterion, we evaluate its correctness. We define an indicator variable $I_m(\text{crit})$ that takes the value of 1 if the criterion successfully identifies the true data-generating copula, and 0 otherwise:

$$I_m(\text{crit}) = \mathbb{I}\left(c^{\text{selected}}_{m,\text{crit}} = c^\circ \right).$$

The empirical hit rate (HR) for a specific criterion, is then calculated as the sample mean of these indicator variables across all M replications:

$$\text{HR}(\text{crit}) = \frac{1}{M} \sum_{m=1}^{M} I_m(\text{crit}).$$

Because the sequence of indicator variables $I_1, \dots, I_M$ constitutes independent Bernoulli trials with a success probability of

$$p = \mathbb{E}\left[\text{HR}(\text{crit})\right] = \text{P}[\text{crit selects } c^\circ],$$

we can construct asymptotic 95% confidence intervals around our estimated hit rates. Let s_{crit} denote the sample standard deviation of the indicator variables $\{I_m(\text{crit})\}_{m=1}^{M}$. Then, by combining the Central Limit Theorem (CLT) and Slutsky's theorem, it follows that

$$\frac{\sqrt{M}\left[\text{HR}(\text{crit}) - p\right]}{s_{\text{crit}}} \xrightarrow[M \to \infty]{\mathcal{L}} N(0, 1).$$

Therefore, 95% confidence intervals are computed using the 0.975 quantile of the standard normal distribution, $z_{0.975} \approx 1.96$. The final reported hit rates in our simulation tables are presented in the format:

$$\text{HR}(\text{crit}) \pm z_{0.975} \times \frac{s_{\text{crit}}}{\sqrt{M}}.$$

Providing confidence intervals for the hit rates allows us to determine whether one criterion performs statistically significantly better or worse than another.

3.3.4 Results and Discussion: Identification of the True Copula

Table 2 presents the hit rates and corresponding 95% confidence intervals for the sample size $n = 200$ and extremely weak dependence $\tau = 0.05$ scenario. This specific regime is challenging in copula model selection. Because the dependence is weak, all candidate copulas are similar to the independence copula, making them difficult to distinguish. A preliminary observation of the table 2 reveals that the identifiability of the true data-generating process varies drastically depending on the true copula family. We see that the Joe copula is the easiest to identify for all criteria, with hit rates ranging from 43% to 53%. For selection criteria ${}^{p}\mathrm{AIC}$, ${}^{p}\mathrm{xv}_1$ and $\mathrm{xv}_{\mathrm{CIC}}$ Clayton copula also shows relatively good identifiability, which is approximately 55%. Clayton and Joe families are heavily asymmetric: the Clayton copula exhibits strict lower-tail dependence, while the Joe copula exhibits strict upper-tail dependence. Even as the overall dependence τ is only 0.05, the functional forms of the Clayton and Joe copulas have asymmetric density concentrations near the corners of the unit square. This structural property may explain the reason why they remain more identifiable than the other models. In general, the ${}^{p}\mathrm{AIC}$, ${}^{p}\mathrm{xv}_1$, and $\mathrm{xv}_{\mathrm{CIC}}$ criteria perform similarly to one another. Furthermore, it appears that the proposed K_n-fold criteria do not significantly outperform the ${}^{p}\mathrm{AIC}$, ${}^{p}\mathrm{xv}_1$, and $\mathrm{xv}_{\mathrm{CIC}}$. The only exception occurs with the Gumbel copula where the most xv_{K_n} criteria perform significantly better. However, it is important to note that the highest hit rate in this scenario is only slightly above 20%. Because this is nearly equivalent to making a uniformly random choice among the five candidate copula families, this specific result can be considered a statistical artifact that does not provide any practical utility.

IC	Clayton	Gumbel	Joe	Frank	Gaussian
${}^{p}\mathrm{AIC}$	55.50 $\pm$3.080	13.20 $\pm$2.100	51.30 $\pm$3.100	32.80 $\pm$2.910	18.00 $\pm$2.380
${}^{p}\mathrm{xv}_1$	55.50 $\pm$3.080	13.20 $\pm$2.100	51.30 $\pm$3.100	32.80 $\pm$2.910	18.00 $\pm$2.380
$\mathrm{xv}_{\mathrm{CIC}}$	55.70 $\pm$3.080	10.60 $\pm$1.910	53.60 $\pm$3.090	29.20 $\pm$2.820	18.40 $\pm$2.400
$K = 2$	40.00 $\pm$3.040	13.80 $\pm$2.140	43.90 $\pm$3.080	23.80 $\pm$2.640	16.20 $\pm$2.280
$K = 5$	38.50 $\pm$3.020	18.00 $\pm$2.380	48.20 $\pm$3.100	22.30 $\pm$2.580	13.80 $\pm$2.140
$K = 10$	38.80 $\pm$3.020	21.50 $\pm$2.550	50.60 $\pm$3.100	25.90 $\pm$2.720	13.90 $\pm$2.150
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	36.70 $\pm$2.990	24.40 $\pm$2.660	52.10 $\pm$3.100	24.60 $\pm$2.670	15.40 $\pm$2.240
$K_n = \sqrt{n}$	38.60 $\pm$3.020	22.40 $\pm$2.590	50.90 $\pm$3.100	24.50 $\pm$2.670	15.40 $\pm$2.240
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	39.90 $\pm$3.040	15.30 $\pm$2.230	43.10 $\pm$3.070	23.70 $\pm$2.640	14.30 $\pm$2.170

Table 2: Hit rates ($n = 200$, $\tau = 0.05$), with 95 % confidence intervals (all values multiplied by 100).

For a larger sample size of $n = 400$ but still relatively weak dependence of $\tau = 0.25$, Table 3 shows that the selection performance of all criteria increased significantly. Overall, the criteria perform similarly, yielding no statistically significant differences among them. The only exception occurs under the Joe, Frank, and Gaussian copulas, where the xv_2 criterion performs worst. The overall trend in the other simulation cases is that increasing either the sample size n or the magnitude of dependence τ will result in an improvement in hit rate performance. From Tables 2 and 3, it is evident that there is generally no statistically significant difference between using fixed or data-driven choices for the number of folds K_n. Therefore, our proposed idea of using data-driven choices for ${}^{p}\mathrm{xv}_{K_n}$ did not result in a superior information criterion. Surprisingly, in most cases, the minimal choice of $K = 2$ performed as effectively as the popular fixed choices of $K = 5$ and $K = 10$. Furthermore, our simulation results confirm the findings of Jordanger and Tjøstheim (2014), specifically that the ^{p}AIC and the $\mathrm{xv}_{\mathrm{CIC}}$ perform almost identically. In comparison to their simulation study, we also computed the ${}^{p}\mathrm{xv}_1$ here. From the two provided tables, one can see that the confidence intervals for the ${}^{p}\mathrm{xv}_1$ and the $\mathrm{xv}_{\mathrm{CIC}}$ overlap for sufficiently large sample sizes ($n = 200$ or $n = 400$). This implicitly confirms the theoretical result that the $\mathrm{xv}_{\mathrm{CIC}}$ provides the correct asymptotic approximation of the ${}^{p}\mathrm{xv}_1$.

IC	Clayton	Gumbel	Joe	Frank	Gaussian
^{p}AIC	97.90 ±0.890	68.70 ±2.880	86.40 ±2.130	75.50 ±2.670	67.80 ±2.900
${}^{p}\mathrm{xv}_1$	97.90 ±0.890	68.70 ±2.880	86.40 ±2.130	75.50 ±2.670	67.80 ±2.900
$\mathrm{xv}_{\mathrm{CIC}}$	96.10 ±1.200	65.90 ±2.940	88.60 ±1.970	77.40 ±2.590	67.00 ±2.920
$K = 2$	96.00 ±1.220	64.70 ±2.960	83.20 ±2.320	69.50 ±2.860	61.90 ±3.010
$K = 5$	97.70 ±0.930	68.20 ±2.890	85.50 ±2.180	75.80 ±2.660	67.50 ±2.900
$K = 10$	97.60 ±0.950	69.40 ±2.860	84.80 ±2.230	75.70 ±2.660	66.90 ±2.920
$K_n = \left\lfloor \frac{n}{\log(n)} \right\rfloor$	97.70 ±0.930	69.20 ±2.860	85.70 ±2.170	77.00 ±2.610	66.90 ±2.920
$K_n = \sqrt{n}$	98.00 ±0.870	68.70 ±2.880	86.20 ±2.140	77.40 ±2.590	67.50 ±2.900
$K_n = \left\lfloor \frac{\sqrt{n}}{\log(n)} \right\rfloor$	97.20 ±1.020	67.90 ±2.900	85.20 ±2.200	75.60 ±2.660	64.90 ±2.960

Table 3: Hit rates ($n = 400$, $\tau = 0.25$), with 95 % confidence intervals (all values multiplied by 100).

3.3.5 Coincidence Percentages

While the hit rate evaluates how frequently a criterion identifies the true data-generating model, it may also be useful to understand how the criteria behave relative to one another. From the previous section, we deduce that in practical applications, it suffices to use the ^{p}AIC, which performs almost identically to the $\mathrm{xv}_{\mathrm{CIC}}$. One might wonder which of the two criteria, leave-one-out cross-validation ${}^{p}\mathrm{xv}_1$ or its approximation $\mathrm{xv}_{\mathrm{CIC}}$, performs more similarly to the ^{p}AIC. To quantify the agreement between the criteria, we define the "coincidence percentage". This metric measures the empirical frequency with which two distinct information criteria select the exact same candidate copula model for a given dataset, regardless of whether that selected copula model is the true data-generating process. Let crit_1 and crit_2 denote two distinct model selection criteria (e.g., $\mathrm{crit}_1 = {}^{p}\mathrm{AIC}$ and $\mathrm{crit}_2 = {}^{p}\mathrm{xv}_1$). As defined in section 3.3.3, let $c^{\mathrm{selected}}_{m,\mathrm{crit}_1}$ and $c^{\mathrm{selected}}_{m,\mathrm{crit}_2}$ represent the specific candidate copula families chosen by criteria crit_1 and crit_2, respectively, during the m-th Monte Carlo replication. We define a coincidence indicator variable $J_m(\mathrm{crit}_1, \mathrm{crit}_2)$ that takes the value of 1 if both criteria select the identical model, and 0 otherwise:

$$J_m(\mathrm{crit}_1, \mathrm{crit}_2) = \mathbb{I}\left(c^{\mathrm{selected}}_{m,\mathrm{crit1}} = c^{\mathrm{selected}}_{m,\mathrm{crit}_2}\right).$$

The empirical coincidence percentage (CP) between the two criteria is calculated as the sample mean of these indicator variables across all M replications:

$$\mathrm{CP}(\mathrm{crit}_1, \mathrm{crit}_2) = \frac{1}{M} \sum_{m=1}^{M} J_m(\mathrm{crit}_1, \mathrm{crit}_2).$$

3.3.6 Results and Discussion: Criteria Agreement

In our simulation study, we evaluate these coincidence percentages across various sample sizes $n \in \{100, 200, 400\}$ and the following values of Kendall's tau $\tau \in \{0.05, 0.1, 0.15, 0.2, 0.25\}$. We intentionally restrict our focus to weak dependence; this represents the most challenging scenario, as distinguishing between competing copula models becomes more difficult. Specifically, we aim to benchmark the ^{p}AIC against the exact ${}^p\mathrm{xv}_1$ and the analytical approximation $\mathrm{xv}_{\mathrm{CIC}}$. We exclude the K_n-fold information criteria ${}^p\mathrm{xv}_{K_n}$ from this analysis because, as demonstrated in the previous section 3.3.4, they offer no performance advantage over the established criteria.

Table 4 shows that while ^{p}AIC is theoretically unjustified and ${}^p\mathrm{xv}_1$ is a theoretically valid information criterion from a predictive perspective, these two criteria are virtually indistinguishable in practice. Across all evaluated sample sizes n and values of Kendall's tau τ, the coincidence percentage remains very high. Even in the most challenging scenario of extreme weak dependence $\tau = 0.05$ and a smaller sample size $n = 100$, the two criteria select the same copula model 99.74% of the time. As the sample size increases to $n = 400$, the coincidence reaches a perfect 100.00% across almost all evaluated levels of Kendall's tau τ. For practitioners, this is a highly valuable result. It provides strong empirical justification that calculating the computationally trivial ^{p}AIC yields results nearly identical to those of the computationally expensive ${}^p\mathrm{xv}_1$.

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	99.74	99.92	99.90	99.96	99.94
200	99.92	99.98	99.98	99.98	99.98
400	100.00	99.98	100.00	100.00	100.00

Table 4: Coincedence of ^{p}AIC and ${}^p\mathrm{xv}_1$ (all values multiplied by 100).

Unlike Table 4, the coincidence percentages in Table 5 exhibit greater variability. From Table 5, one can see that for smaller sample sizes n, there is a lower coincidence rate between ^{p}AIC and $\mathrm{xv}_{\mathrm{CIC}}$. Because Table 4 demonstrates that ^{p}AIC performs almost identically to ${}^p\mathrm{xv}_1$, the discrepancy in Table 5 suggests that $\mathrm{xv}_{\mathrm{CIC}}$ provides a poorer approximation of ${}^p\mathrm{xv}_1$ in small samples. This is expected, as the bias-correction terms required to compute $\mathrm{xv}_{\mathrm{CIC}}$ are less stable for smaller sample sizes. However, as seen in Table 5, as the sample size increases, the $\mathrm{xv}_{\mathrm{CIC}}$ penalty stabilizes and behaves more similarly to ^{p}AIC. In other words, as the sample size n increases, $\mathrm{xv}_{\mathrm{CIC}}$ becomes a more accurate approximation of ${}^p\mathrm{xv}_1$.

	$\tau = 0.05$	$\tau = 0.1$	$\tau = 0.15$	$\tau = 0.2$	$\tau = 0.25$
100	79.26	86.46	89.10	89.34	90.34
200	86.88	91.38	92.90	94.10	94.72
400	92.52	95.46	96.38	96.60	97.26

Table 5: Coincedence of ^{p}AIC and $\mathrm{xv}_{\mathrm{CIC}}$ (all values multiplied by 100).

4 Conclusion

Selecting the most appropriate copula model to capture the dependence structure between random variables is a fundamental challenge in multivariate statistics. This project investigated this problem through two distinct methodological approaches: the fully parametric approach utilizing two-stage maximum likelihood estimation (2MLE), and the semiparametric approach utilizing maximum pseudo-likelihood estimation (MPLE). By independently replicating and a little extending key simulation studies from the literature, we evaluated established information criteria.

In the fully parametric setting, we replicated the simulation study by Ko and Hjort (2019) to compare the performance of $\mathrm{AIC}_{2\mathrm{MLE}}$ and the model-robust $\mathrm{CIC}_{2\mathrm{MLE}}$. Our results confirm that while $\mathrm{CIC}_{2\mathrm{MLE}}$ penalizes misspecified models more heavily through its complex bias-correction term, this conceptual superiority provides little practical advantage for large sample sizes. This occurs because the shared log-likelihood term dominates the respective penalty terms, resulting in nearly identical model rankings between the two criteria.

In the semiparametric setting, we evaluated criteria in which the marginal distributions are estimated nonparametrically. Replicating Jordanger and Tjøstheim (2014), we compared the theoretically unjustified ${}^p\mathrm{AIC}$ against the rigorously derived leave-one-out cross-validation ${}^p\mathrm{xv}_1$ and its asymptotic approximation $\mathrm{xv}_{\mathrm{CIC}}$. Our extensive simulations demonstrate that the ${}^p\mathrm{AIC}$ performs almost indistinguishably from the theoretically valid ${}^p\mathrm{xv}_1$. Furthermore, our results empirically validate the asymptotic derivations of Grønneberg and Hjort (2014), showing that $\mathrm{xv}_{\mathrm{CIC}}$ serves as an excellent approximation of xv_1 in large samples. In smaller samples, however, $\mathrm{xv}_{\mathrm{CIC}}$ can deviate from ${}^p\mathrm{xv}_1$ because the bias-correction terms required for its computation introduce finite-sample variation. Additionally, we proposed and evaluated a K_n-fold cross-validation criterion, ${}^p\mathrm{xv}_{K_n}$. The motivation was that employing data-driven choices for the number of folds would mimic the asymptotic properties of ${}^p\mathrm{xv}_1$ and would be less computationally expensive; thus, we investigated how such a criterion would perform. The simulation results demonstrate that data-driven dynamic folds K_n do not yield a statistically superior information criterion compared to existing criteria.

Ultimately, this whole project demonstrates that choosing a model selection technique in the copula context depends heavily on the specific application. Let's say, for example, that our application is in a high-dimensional setting, meaning we cannot use classical criteria like $\mathrm{AIC}_{\mathrm{MLE}}$ or $\mathrm{TIC}_{\mathrm{MLE}}$. The first question one should ask is whether the true nature of the marginal distributions is known—perhaps from domain experts. If it is known, then one should use a fully parametric strategy with two-stage MLE. If not, one is better off sticking to the semiparametric strategy and MPLE. Once the strategy is chosen, the next step is deciding which information criterion to apply. Suppose we are in the parametric context deciding between $\mathrm{AIC}_{2\mathrm{MLE}}$ and $\mathrm{CIC}_{2\mathrm{MLE}}$. If you have a very large sample size, you can safely use $\mathrm{AIC}_{2\mathrm{MLE}}$. As our simulation study indicates, the two criteria perform almost identically, regardless of whether the true model is misspecified. However, if the sample size is small and the model is misspecified, one should still be careful with $\mathrm{CIC}_{2\mathrm{MLE}}$; since it was derived as an asymptotic approximation, it implicitly assumes a large n is available to stabilize its bias-correction terms. A similar discussion applies to the semiparametric approach when choosing between ${}^p\mathrm{AIC}$ and $\mathrm{xv}_{\mathrm{CIC}}$. But here, there is a major contrast to the parametric framework. In the parametric case, $\mathrm{AIC}_{2\mathrm{MLE}}$ is a theoretically valid criterion, so we mathematically understand why it behaves similarly to $\mathrm{CIC}_{2\mathrm{MLE}}$ for large samples. On the other hand, ${}^p\mathrm{AIC}$ lacks theoretical justification. Why it performs so similarly to exact leave-one-out cross-validation ${}^p\mathrm{xv}_1$ in simulations remains an interesting finding, which is not entirely clear from a theoretical point of view.

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